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PALEOMAGNETISM OF SAMPLES FROM THE AXIAL ZONE OF THE AFAR DEPRESSION *

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A series of oriented lava flows has been collected for paleomagnetic work from an axial graben situated in the Afar depression. This graben has many geophysical and structural features which suggest a similarity with the median valleys found in many mid-oceanic ridges. The results of the paleomagnetic work show that all the samples collected from the graben have normal magnetizations. This, coupled with the fact that they are obviously very young, suggests that they were emplaced during the Brunhes normal epoch of the earth's magnetic field. A series of four older lavas was sampled. These lavas gave directions just below the horizontal, but with easterly declinations. These directions could have been caused by a deviation of the earth's field from its normal axial dipole direction, or they could be caused by large-scale rotation about a vertical axis. This rotation could be connected to the distortion necessary to close up the continental portions of crust around the Gulf of Tadjoura.

1. Introduction

The Afar depression, situated in Ethiopia and Territoire Française des Afars et des Issas (T.F.A.I.) is a region of recent volcanic activity. It has been compared to a spreading center and it is postulated that spreading in the Afar depression during the Neogene has caused the separation of the Danakil horst to the west of the depression from the main Ethiopian Plateau to the east. It is a region of great interest as it is the junction of several major tectonic provinces, the Red Sea spreading center, the Gulf of Aden spreading center and the East African rift valley. The spreading center in the Gulf of Aden appears to continue into the Gulf of Tadjoura and thence into Ghoubbat-al-Kharab [1]. From there the spreading center continues on land and forms the graben responsible for Lake Asal. Spreading is postulated to terminate in the region of Lake Asal, and to continue in another spreading segment of recent vulcanism to the northeast, the two spreading segments being joined together by the continental equivalent of a transform fault [2].

There is abundant geophysical evidence that the Afar depression has many similarities to recent spreading centers in the ocean, although to some extent its structure is intermediate between truly continental and truly oceanic. Ruegg [3] has shown that in the seismic layering around the Gulf of Tadjoura in T.F.A.I. anomalously low mantle velocities are found at rather shallow depths, and the seismic structure in this region looks very similar to that in Iceland or close to the Mid-Atlantic Ridge. It is definitely not like normal continental structure. Berktold et al. [4] have shown that magnetotelluric measurements made in the depression suggest anomalously high temperatures which give a temperature gradient of about 60°C/km. Seismic work in the main part of the depression by Berckhemer et al. [5] demonstrates clearly that Afar is approaching an oceanic type of structure. One of the most convincing pieces of evidence is that described by Searle [6] who studied the dispersion of surface waves travelling across the depression. He found that

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most models gave a P-wave velocity of over 7 km/s (anomalous mantle) at depths of about 15–17 km, thus making Afar intermediate in structure between continental type and oceanic type. Even the surficial features of the area in which the paleomagnetic study was made, between Ghoubbat-al-Kharab and Lake Asal, have many similarities to mid-oceanic median valleys [1,7]. Volcanic activity in the area under investigation is of fissural basalts, intermediate between the alkali basalts on either side of the axial graben and true oceanic tholeiites [8,9].

Since it is difficult to obtain oriented samples from oceanic spreading centers for paleomagnetic work, we decided to collect oriented samples from the axial region between Ghoubbat-al-Kharab and Lake Asal in order to make paleomagnetic measurements.

2. Location of the samples

The locations of the oriented samples are shown in Fig. 1. Except for lavas 29–32, which come from older trap volcanics away from the axial valley, all the samples were taken within the axial valley in lavas which are described as recent to subrecent intermediate basalts [8]. In all, thirty-two individual lavas were sampled, of which four were from the older trap series. In most cases, two cores were drilled from each lava flow, using a gasoline-powered drill. The cores were oriented using a magnetic compass and a sun compass, except for lavas 17–19, which were collected on the north side of a steep cliff where the sun did not shine in winter. These were oriented using a magnetic compass only. The agreement between the sun compass and magnetic compass for the other samples was good (r.m.s. deviation of 1.6°). We therefore assume that the magnetic compass orientations of lavas 17–19 are correct to within a few degrees.

3. Magnetization directions

The directions of magnetization were determined using a Foster-type spinner magnetometer. All samples were demagnetized using alternating magnetic induction techniques. Rather than rely on

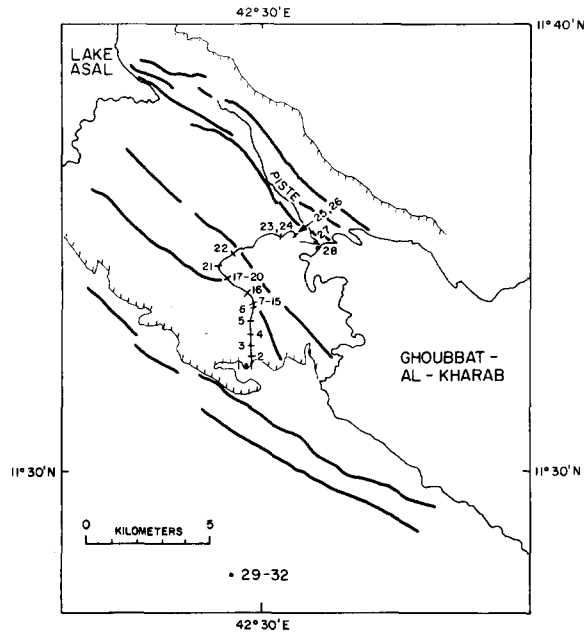


Fig. 1. Map of location of lava flows. The shaded lines show the boundary between the younger intermediate lava flows with phenocrysts, and hyaloclastites, between the lines, and the older alkali basalts towards the NE and SW. Samples 29–32 come from the lower trap series. The shorter heavy lines show the locations of the more prominent dip-slip faults.

determining the demagnetizing induction which best removes the secondary components of magnetization from one sample and then applying this one demagnetizing induction to other samples from the same flow, we have run each sample through a complete demagnetization process and have used the method of Tarling and Symons [10] to analyze the results. This method depends on determining the maximum value of the parameter:

$$S = R^{1/2}/\theta_{63^\circ}$$

for three or more observations of the direction of magnetization, and after two or more successive increases in the demagnetizing induction. The parameter R is the range in induction values (measured in gauss or 10^{-4} tesla) and θ_{63° is the circular standard deviation of the directions of magnetization [11].

The maximum value of S is the stability index as defined by Tarling and Symons [10]. The directions of magnetization which give the maximum value of S

are all tightly grouped in this set of samples, as the stabilities are high. Rather than arbitrarily choosing one of the directions of magnetization within the range of three or more to represent the stable direction of magnetization, we have chosen to average together all the directions of magnetization which are involved in the determination of the maximum value of S. There would have been almost no difference if only one of these directions had been chosen to represent the stable direction of magnetization. The advantage of using this method of determination of a stable direction of magnetization is that it is completely objective. Once the demagnetization has been done, the direction chosen as stable is fixed. Directions can only be thrown out if either the stability index is low, meaning that the directions do not group well, or if there is an inconsistency between different samples from the same lava flow.

Fig. 2 shows a histogram of the stability indices established for all of the samples studied. Samples with values of the stability index greater than 2.5 were classified as "stable" by Tarling and Symons [10], with classifications of "very stable" for values greater than 5 and "extremely stable" for values greater than 15. It can be seen that all the samples were stable under this classification scheme, and 89% were very stable or extremely stable.

In most cases, the directions of magnetization did not change much during demagnetization. This is illustrated in Fig. 3, which shows the directions of magnetization obtained from sample 24 during demagnetization up to a maximum peak induction of 5.77×10^{-2} T. This sample gave a stability index of 40, for directions after demagnetization at 2.08, 3.46 and 5.77×10^{-2} T. Other samples gave large changes in the direction of magnetization during demagnetization. One such sample was 19B and the demagnetization curve for this sample is also shown in Fig. 3. This sample had a high NRM intensity of 10.6 A/m, which was quickly reduced during demagnetization, such that the median destructive induction (i.e., that necessary to reduce the magnetization intensity to one-half its NRM value) was only 7.3×10^{-3} T. The stability index was established for the three directions after demagnetization at 2.08, 3.46 and 5.77×10^{-2} T, and was 15.4. Another sample which showed a large change in the direction of magnetization during demagnetization was 7 (Fig. 3). In this case, the stability index was only 3.7, established for the last three directions, and for the same demagnetizing inductions as in the other two cases shown in Fig. 3.

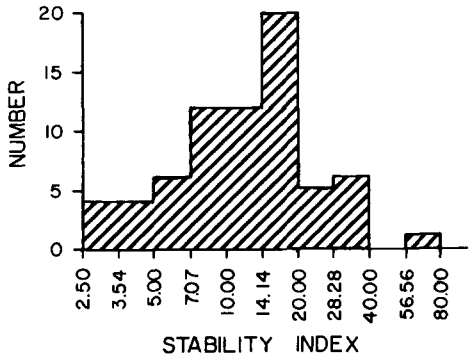


Fig. 2. Histogram of stability indices. Index calculated using the method of Tarling and Symons [10].

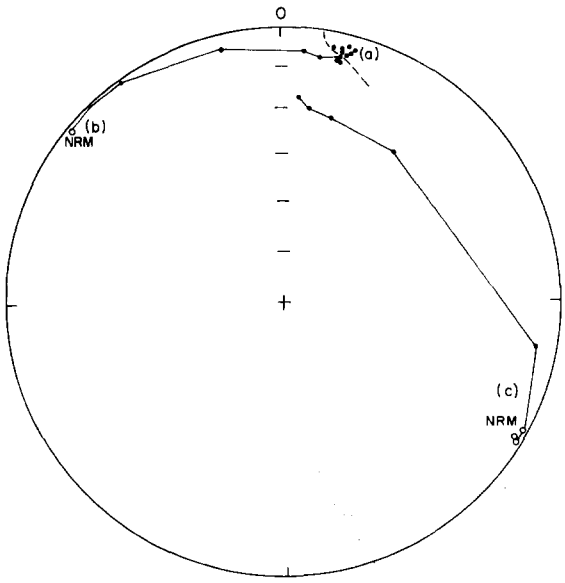


Fig. 3. Directions of magnetization during demagnetization. (a) Sample 24. NRM plus seven demagnetized directions up to a maximum induction of 5.77×10^{-2} T. (b) Sample 19B. Open circle shows negative inclination. (c) Sample 7.

These results show that the demagnetizing technique was working satisfactorily, and that the samples

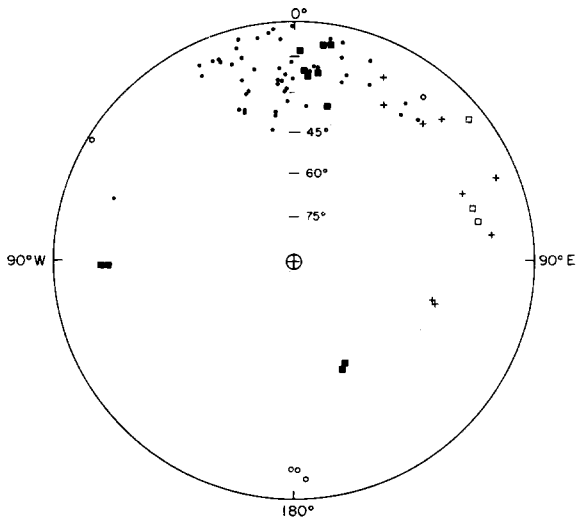


Fig. 4. Equal area plot of NRM directions. The solid symbols show directions below the horizontal. The open symbols show directions above the horizontal. The crosses show directions from the four older lava flows (nine samples); these directions were all below the horizontal, i.e., with positive inclinations. The squares are results from samples for which there was no internal consistency. In other words, samples from the same lava flow gave very different directions of magnetization, which could not be made to agree by demagnetization. There were four lava flows and a total of fourteen samples in this category.

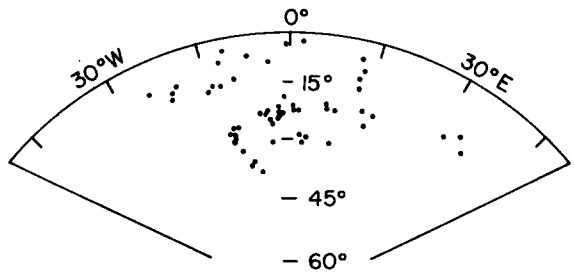


Fig. 5. Equal area plot of directions of magnetization after partial demagnetization. These directions were from the 24 young lava flows which had internal consistency.

had in general a high degree of stability towards alternating induction demagnetization. Most samples had not acquired much secondary magnetization. Because of the high degree of stability, none of the samples was rejected from statistical analysis on the basis of lack of stability.

The directions of magnetization before demagnetization (NRM directions) are shown in Fig. 4. When demagnetization was done, the samples could be separated into three different groups. One group, consisting of 55 samples from 24 lava flows, had good internal consistency, in that all samples from each lava flow agreed. One group, consisting of fourteen samples from four lava flows, did not show good internal consistency. This may be caused by errors in orientation, or possibly remagnetization by lightning. The third group, consisting of nine samples from four lava flows, gave internally consistent directions of magnetization, but this direction of magnetization was very different from the direction of magnetization from the first group of twenty-four lava flows. These four lava flows were the older ones sampled to the southwest of the axial zone (Fig. 1.).

Directions of magnetization after partial demagneti zation are shown in Fig. 5 for the first group of twenty-four lavas. All samples have positive inclinations ranging from almost horizontal to 40°. The decli nations scatter 25–30° on either side of true north. Directions of magnetization from the four older lava flows are shown in Fig. 6. The cross is the mean of these nine directions of magnetization. Seven of these nine samples are classified as "very stable" or "ex-

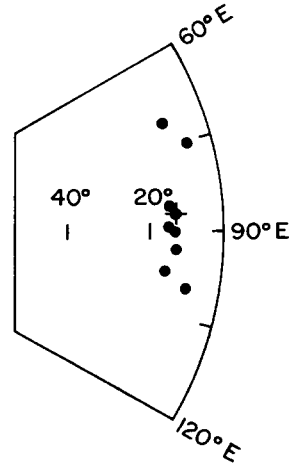


Fig. 6. Equal area plot of directions of magnetization after partial demagnetization of samples from the four older lava flows. The cross is the mean direction of magnetization calculated according to the method of Fisher [16].

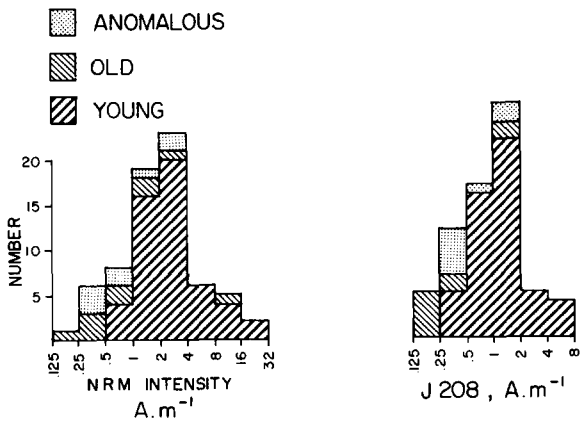


Fig. 7. Histogram of intensities of NRM and intensities of magnetization after demagnetization in an induction of 208×10^{-4} T. The results are separated into the three groups discussed in the text.

tremely stable" according to the scheme of Tarling and Symons [10]. The declinations are easterly and the inclinations are all less than 20°.

4. Intensities of magnetization

Intensities of magnetization were measured, and are shown as a histogram in Fig. 7. In this histogram, the younger lavas, the older lavas and the anomalous lavas have been shown separately. As intensities of remanence tend to be distributed log-normally [12], we have calculated the geometric mean and standard error for these three groups of rocks. These are shown in Table 1. Also included is the mean and standard error for the three samples of hyaloclastite, which were not included in Fig. 7. We have also calculated the intensity of magnetization after demagnetization in a peak induction of 2.08×10^{-2} T, and these values are shown in Table 1 and Fig. 7. All the values in Table 1 may be compared with those for subaerial lavas given by Prévot and Grommé [13]. The young volcanics have a mean magnetization which is within the error limits of the mean magnetization of the historic flows of Prévot and Grommé, but is lower than the mean value for lavas between 0.01 and 1.0 m.y. old. However, comparing the values for demagnetization in an induction of 2.08×10^{-2} T (this paper) with demagnetization in 2×10^{-2} T (Prévot and Grommé), the value for the younger lavas is well within the error limits for the lavas between 0.01 and 1.0 m.y. old.

The older volcanics from T.F.A.I. have a significantly lower intensity of magnetization than the younger lavas from the same area, and are also much lower than the mean values given by Prévot and Grommé. However, approximately 10% of the lavas listed by Prévot and Grommé have intensities which, when reduced to the equator, are less than those of the older volcanics from T.F.A.I. The demagnetized

TABLE 1

Magnetization intensities

Group	$\bar{J}_{NRM}$ (A/m)	Standard error of mean (A/m)	Number of samples	$\bar{J}_{208}$ (A/m)	Standard error of mean (A/m)
Young volcanics	2.737	+0.276 −0.268	52	1.266	+0.137 −0.124
Old volcanics	1.123	+0.577 −0.381	9	0.279	+0.086 −0.065
Anomalous volcanics	0.919	+0.375 −0.266	14	0.536	+0.165 −0.126
Hyaloclastites	4.396×10^{-2}	$+0.534 \times 10^{-2}$ -0.477×10^{-2}	3	2.349×10^{-2}	$+0.243 \times 10^{-2}$ -0.209×10^{-2}

$\bar{J}_{208}$ is the average magnetization after demagnetizing in an induction of 208×10^{-4} T.

results from the older volcanics fall further outside the distribution of Prévot and Grommé than do the NRM results. The same may be said about the anomalous results from Afar, except that in this case the NRM intensities are further outside the distribution of Prévot and Grommé than are the demagnetized intensities.

However, the values of intensities for the anomalous results would not be cause to exclude these results from further statistical analysis. The exclusion rests solely on the anomalous directions between samples taken from the same lava flow. This is not true for the older samples, where samples taken from the same lava flow agree to within a few degrees.

5. Statistical analysis of young lava flows

There are not enough samples to allow us to do a good statistical analysis of the directions of magnetization in order to measure the secular variation component of the magnetic field in the region. For instance, Doell [14] in his study of the secular variation in the Massif Central, France, used up to eight samples from each lava flow, and analyzed thirty-one lava flows of Brunhes age in order to determine the secular variation for this place and time Ellwood et al. [15] have suggested a minimum number of samples of three or four from each lava flow in order to arrive at a good estimate of the secular variation. It would have been impossible to sample laterally over the extent that Doell did in the Massif Central (up to 177 m) as the exposed portions of each lava were not extensive enough. We sampled twenty-eight young lava flows, of which four were rejected because of lack of internal consistency; unfortunately some of the lava flows were only sampled once. The average number of samples from each lava flow was just over 2 (2.29), which means that an insufficient

TABLE 2

Directions of magnetization in 24 lavas, after partial demagnetization

Lava No. (see Fig. 1)	Declination (°E)	Inclination (°down)	Number of samples	Resultant vector	α_{95}°	Angular standard deviation (°)
1	14.9	21.7	2	1.9992	7.0	1.6
2	20.8	17.6	4	3.7540	27.4	20.2
3	348.5	31.4	1			
4	352.6	21.8	4	3.8635	20.0	15.0
5	359.9	2.5	2	1.9991	4.1	1.7
6	357.9	22.4	4	3.9992	2.3	1.2
7	2.9	26.5	2	1.9906	24.6	5.6
8	355.1	25.9	2	1.9949	18.0	4.1
10	349.4	12.1	1			
12	355.8	22.4	2	1.9985	9.8	2.2
14	356.4	26.1	2	1.9993	6.7	1.5
15	352.2	17.5	4	3.8951	17.5	13.2
16	351.3	35.4	1			
17	0.6	20.8	2	1.9999	3.3	0.6
19	13.3	12.3	2	1.9990	7.9	1.8
20	10.7	19.9	2	1.9979	11.1	2.6
21	11.8	4.9	1			
22	6.0	20.1	2	1.9999	2.1	0.6
23	15.0	14.2	2	1.9998	2.2	0.9
24	12.7	8.5	1			
25	338.3	12.1	2	1.9997	3.5	11.0
26	337.1	9.5	2	1.9983	10.2	22.4
27	347.9	26.9	5	4.9994	1.7	0.9
28	6.3	23.1	3	2.9850	10.7	5.7

TABLE 3

Analysis of dispersion – young lavas

Source	Degrees of freedom	Sum of squares	Mean square
Between sites	46	1.521251	0.0330707
Within sites	62	0.318252	0.0051331
Total	108	1.839503	

number was obtained to do an adequate analysis of secular variation. For this reason, we have only analyzed directions of magnetization. For secular variation studies, it is now customary to analyze pole positions.

Table 2 shows mean values of direction of magnetization in the twenty-four lava flows after partial demagnetization. Also shown are the number of samples collected from each lava flow, the resultant vector for the directions in each lava flow, the radius of the circle of confidence around the mean direction (α_{95}) and the angular standard deviation of directions from the mean direction. These last three results are of course not included when only one sample was available for a lava flow. The methods of Fisher [16] and Wilson [17] were used to obtain these numbers.

In order to obtain a mean direction of magnetization from all 24 lava flows, and to estimate the error of this mean direction, it is necessary to allow for the variation in direction associated with individual lava flows. This is done by using the two-tier analysis which is described by McElhinny [18]. The results are given in Table 3. The between-site variation is significant, as may be seen by dividing the second number in the last column into the first number and looking up in a table of the F distribution. Therefore we can calculate the Fisher precision parameter for within-site dispersion as being $k_w = 97.41$ and the between-site dispersion parameter as being $k_b = 40.60$. The mean of all 110 directions will be distributed with a precision parameter of $k_0 = 824.5$, which gives a confidence circle around the mean direction of 4.9° at the 95% level. The mean of all 110 directions is given in Table 4 along with axial and present field directions. These directions are also shown in Fig. 8. It can be seen that the average direction is very close to that expected from a central axial dipole field. The mean direction has an inclination which is 3.1° shallower than the axial field. If a pole position is calculated from this direction of magnetization, we obtain a value of 88.0°N, 254.7°E with a cone of confidence having semi-axes of 2.7° and 5.1°. This pole is on the far side of the axis of rotation of the earth, although not significantly different from it. It thus seems to support the hypothesis of Wilson [19] that the field of the earth is better approximated by an axial dipole shifted slightly towards the north pole from the center of the

TABLE 4

Directions of magnetization

	Inclination (°down)	Declination (°E)	k_b	k_w	α_{95}°
24 young flows	19.2	358.9	41	97	4.9
4 old flows	12.0	87.3	101	198	7.7
Axial field	22.3	0.0			
Present field	4.7	359.8			

TABLE 5

Directions of magnetization in 4 older lavas, after partial demagnetization

Lava No. (see Fig. 1)	Declination (°E)	Inclination (°down)	Number of samples	Resultant vector	α_{95}°	Angular standard deviation (°)
29	73.6	8.9	2	1.9965	14.8	3.4
30	93.4	14.2	2	1.9963	15.3	3.5
31	93.3	12.5	1			
32	88.9	12.0	4	3.9820	9.3	5.4

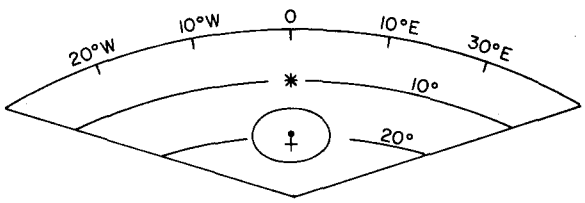


Fig. 8. Mean direction of magnetization for 24 young lava flows, with circle of confidence around mean direction. Also shown is the field direction produced by an axial geocentric dipole (plus), and the present field direction (asterisk).

earth. The northward shift for the Afar data is 126 km. This is somewhat less than the average offset found by Wilson.

The four older lava flows give a very different mean direction of magnetization. Table 5 presents the results of demagnetization for these four lava flows, and in Table 6 is given details of a two-tier analysis of dispersion. One surprising feature of the analysis of dispersion is the close grouping of directions from different lavas, as shown by the large value of k_b. Values of k which are derived from suites of samples which are thought to have adequately recorded the secular variation are in the region of 20–40, whereas the value of k_b for the older lavas is over 100. If we convert the value of k_b into a VGP angular standard deviation using the methods given by McElhinny and Merrill [20], a value of 6.6° is obtained for S, which is much lower than any value given in table 3 of McElhinny and Merrill [20]. This suggests that these lavas were extruded within a fairly short period of time. Two possible explanations can be thought of for the anomalous direction of magnetization. The first is that the field was recorded during a magnetic excursion or during a period of time when the field was in the process of reversing itself. In this case, the lava flows would have to have been extruded within a time period of probably less than 100 years. If they were extruded over a time period longer than this, then the individual lavas would probably have shown some coherent change in direction of magnetization, because during the process of a reversal, or during an excursion, the field probably changes radically in direction over a time of 1000 years. The second possibility is that the lavas originally recorded a field which was normal or reversed, and that subsequently the area underwent a large amount of tectonic disturbance, with the result that the lavas were twisted about a vertical axis by about 90°.

TABLE 6

Analysis of dispersion – old lavas

Source	Degrees of freedom	Sum of squares	Mean square
Between sites	6	0.07661	0.012768
Within sites	10	0.02520	0.002520
Total	16	0.10181	

6. Discussion

It seems reasonable to suppose that all the younger lavas were extruded during the Brunhes normal epoch of polarity. Since the direction of spreading here is approximately N35°E, and since flow 28 (Fig. 1) was about at the axis of the median valley, the total coverage of Brunhes age material, assuming symmetric spreading, will be greater than about 11 km. Delibrias et al. [21] have estimated the spreading rate of about 11 mm/year; if the rotation pole and rate of opening between Africa and Arabia of Minster et al. [22] is used to calculate a spreading rate at the Asal rift, a figure of 9 mm/year is obtained, both these values being for one side of the spreading center only. Thus, during the Brunhes epoch a total amount of 9.8–12.0 km of normally magnetized crust should have been produced. We thus estimate that the lavas sampled further away from the axial zone were close to the edge of the Brunhes age material.

We hope that we shall be able to map the location of the first reversal, from Brunhes normal polarity to Matuyama reversed polarity, by collecting samples from lava flows further away from the axial region. However, it is encouraging that no reversely magnetized lavas were found in the region already under study. This is in contradistinction to results from the FAMOUS area of the Mid-Atlantic Ridge, where many apparently reversely magnetized samples were collected by submersible in the axial zone, which should have been normally magnetized according to the beliefs of sea-floor spreading. Other studies of

the axial zone of the Mid-Atlantic Ridge have also recovered reversed magnetized lavas [23].

The peculiar results from the set of older lavas obviously require further study. Two possible causes for these results can be suggested. One is that they represent an excursion of the earth's magnetic field. Excursions have been observed in lava flows several times before. These excursions are presumed to be times when the axial dipole strength became smaller either in relation to the non-dipole field or in relation to equatorial dipoles. Sometimes such a situation leads to a reversal, and intermediate directions of the field have been observed in lava flows between sections of normal and reversed polarity.

Doell [24] has documented a case of an excursion seen in the lavas exposed on the Snaefellsnes peninsula of Iceland, in which eight consecutive lavas gave anomalous pole positions. However, there were no groupings of four consecutive directions of magnetization which were anything like as close as the grouping shown by the older lavas from Afar. Doell and Dalrymple [25] have studied paleomagnetic directions from the Waianae and Koolau volcanic series from Oahu and have discovered three excursions of the field. Two of the excursions are recorded in only one flow each, but the third is recorded in two flows. In each case involving one flow, there is either a flow beneath or a flow above which has directions of magnetization consistent with a predominantly axial dipole field. In the case involving two flows, there are flows both above and below which have normal directions of magnetization.

Another case of a possible excursion is different from the ones described above. It involves lava flows collected from the island of Necker [26]. All seven flows collected from this island had anomalous directions, which were in two different groups. One of the groups had five directions which differed by less than 15°. Doell considered these to be caused by anomalous directions of magnetization, and not by a tectonic effect. But it is certainly different from the results seen in the other excursions. Either the lavas from Necker Island were erupted during a very short period of time compared to lava flows recording the other excursions, or the anomalous direction of the magnetic field was constant for a much longer period of time.

The second possible cause for the easterly directions of magnetization observed in the older lava flows is the possibility that they are the result of rotation of a large block about a vertical axis, which would give large changes in declination, but little change in inclination. Obviously more observations of easterly or westerly declinations are necessary before this hypothesis can be proved. It is, however, interesting to note that the Aisha horst, a northward projection of the continental Somali Plateau lying between 43°E and 44°E, and shown on maps in Barberi et al. [27] and Black et al. [28], cannot be kept in its present position if the Red Sea and Gulf of Aden are closed. It is possible that the Aisha horst has undergone rotation. Although the older volcanics described in this paper are not part of the Aisha horst, it is possible that they may have become involved in such a postulated rotation.

7. Comparison with other paleomagnetic work in Afar

There have been other paleomagnetic investigations of very young rocks from the Afar depression. As is to be expected, none of these results shows any significant deviation from an axial field direction. References to these studies may be found in Schult [29]. Older rocks from the region (for instance, the Aden volcanics, studied by Tarling, [30]; Miocene and Pliocene rocks from the Danakil horst, studied by Schult [31] and Burek [32]) begin to show deviations from axial field directions which are usually interpreted as being caused by rotations of the various blocks. For instance, it is usually assumed that Arabia has rotated counterclockwise with respect to Nubia (the name for the plate to the north and west of the African rift valley). This counterclockwise rotation formed the Red Sea. Likewise, the Danakil horst is thought to have rotated counterclockwise with respect to Nubia, creating the Afar depression. The amount of rotation for the Danakil horst must have been more than the amount of rotation suffered by Arabia, and also the rotation pole must be closer for the Danakil horst than for Arabia, otherwise unacceptable geometries are obtained. Such rotations are acceptable from the point of view of paleomagnetic data, as the pole position for the Danakil horst is significantly different from those associated with Arabia [29].

8. Conclusions

(1) Young volcanics from the axial graben between Ghoubbat-al-Kharab and Lake Asal have normal magnetizations. These volcanics are assumed to have erupted during the Brunhes normal epoch of the earth's magnetic field.

(2) The directions of magnetization have a scatter which is consistent with deviations in magnetic field produced by the secular variation. However, the number of samples collected was insufficient to obtain an accurate measurement of the angular deviation in the directions of magnetization.

(3) A series of four older flows gave a mean direction which deviated significantly from an axial dipole field. This may be due either to anomalous field directions during the course of a reversal, or to an excursion of the field, or possibly to tectonic rotation. Further work is necessary to elucidate this problem.

(4) The intensities of magnetization of the lavas sampled in Afar are similar to magnetization intensities of the subaerial lavas discussed by Prévot and Grommé [13]. A predominant number of the Afar lavas was probably erupted subaerially, as submarine lavas of comparable age studied by Prévot and Grommé have much higher intensities. As very high intensities have not been measured, the problem of remagnetization by lightning seems to be unimportant in this area.

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